

**DAVID ZHANG**  
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### **Academic Appointments**

**Assistant Professor of Finance**, 2022-Present (on leave 2026-2027)  
 Jones Graduate School of Business, Rice University, Houston, TX

**Visiting Assistant Professor of Finance**, 2026-2027  
 Kellogg School of Management, Northwestern University, Evanston, IL

### **Education**

**Ph.D. in Business Economics**, Harvard University, 2016-2022  
 Committee: John Campbell (chair), Adi Sunderam, Edward Glaeser, Ariel Pakes, Robin Lee  
 Fields: Real Estate Finance, Household Finance, Industrial Organization

**B.A. in Economics and Mathematics**, Amherst College, 2009-2013, *summa cum laude*

### **Teaching Experience:**

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|------------------|------------------------------------------------------------------------------------------|
| 2022-            | MGMT 648 MBA Applied Finance, Instructor                                                 |
| 2023-2024 Spring | MGMT 746 MBA Real Property, Instructor                                                   |
| 2020 Spring      | Ec 2727 Empirical Methods in Finance, Teaching Fellow for Samuel Hanson and Adi Sunderam |
| 2018 Summer      | APMA S-115 Mathematical Modelling, Teaching Fellow for Zhiming Kuang                     |
| 2018 Fall        | Ec 2610 Industrial Organization I, Teaching Fellow for Ariel Pakes and Robin Lee         |
| 2017 Fall        | Ec 2610 Industrial Organization I, Teaching Fellow for Ariel Pakes and Robin Lee         |

### **Other Experience:**

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|-----------|-------------------------------------------------------------------------|
| 2024-     | Visiting Scholar at the Federal Reserve Bank of Atlanta                 |
| 2019-2024 | Visiting Scholar at the Federal Reserve Bank of Boston                  |
| 2019-2020 | Research Assistant to Professors Ariel Pakes, Mark Shepard, and Kate Ho |
| 2017      | Research Assistant to Professor Adi Sunderam                            |
| 2014-2016 | Research Assistant at the Federal Reserve Bank of Boston                |
| 2013-2014 | Junior Economist at Legal Economics                                     |

### **Professional Activities:**

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|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Invited presentations: | 2026<br>Conferences: MFA (presentation, 2x discussant), Kentucky Finance Conference, NUS-Baruch Real Estate Conference, Philadelphia Fed Mortgage Markets Conference, SFS (2x discussant) <sup>s</sup> , IU & Chicago Fed Real Estate and Urban Policy Symposium (discussant) <sup>s</sup> , Boulder Consumer Finance Conference <sup>s</sup> , AREUEA National Conference (discussant) <sup>s</sup> , Pre-WFA Real Estate Conference <sup>s</sup> , WFA <sup>s</sup> .<br>Seminars: Baruch (Real Estate) <sup>s</sup> . |
|                        | 2025<br>Conferences: AEA, AREUEA/ASSA (discussant), MFA (presenter, discussant), Federal Reserve Bank of Philadelphia, Georgia Tech-Atlanta Fed Consumer Credit Conference <sup>*</sup> , Pre-WFA Real Estate Conference <sup>*</sup> , WFA, SFS (discussant), NBER Summer Institute Real Estate <sup>*</sup> , NFA, UNC Junior Finance Conference (discussant), Holden Conference on Real Estate and Finance (discussant), 4th CEMLA / Dallas Fed Financial Stability Workshop.                                         |

Seminars: Baylor University (Economics), Dartmouth College (Tuck), HKU (Finance), University of Houston (Finance).

#### 2024

Conferences: AEA, CICF (discussant), Cornell Economics Alumni Conference, Philadelphia Fed Mortgage Markets Conference, AREUEA National Conference, MoFiR Conference on Banking, UCLA/SF Fed Conference on Housing, Financial Markets and Monetary Policy (discussant), Northeastern Finance Conference\*, University of Washington Fostering Inclusion Workshop\*, PUC-Chile Virtual Seminar, Federal Reserve Bank of Atlanta Seminar, MIT GCFP 11th Annual Conference\*, University of Toronto Financial Economics Conference, National University of Singapore (Finance), Colorado Finance Summit.

Seminars: University of Iowa (Tippie), University of Rochester (Simon), University of Colorado Boulder\*, Georgetown University (McDonough).

#### 2023

Conferences: AREUEA/ASSA (2x), Penn State (Macro), UCLA/SF Fed Conference on Housing, Financial Markets and Monetary Policy, Boulder Conference on Consumer Finance, SFS Cavalcade North America (discussant), AREUEA National Conference (discussant), North American Summer Meeting of the Econometric Society (NAMES), Western Finance Association (WFA), Northern Finance Association (discussant), Lone Star Finance Conference, FMA\*. Seminars: CityUHK, Zillow, FHFA.

#### 2022:

Conferences: CFPB Research Conference, Chicago Booth Household Finance Conference, NBER Summer Institute Real Estate Conference, UEA Conference (presenter, discussant), North America Summer Meeting of the Econometric Society (NAMES), AREUEA National Conference, Chicago Fed Conference on Risk and Racial Bias (discussant), Asian Meeting of the Econometric Society, OSU PhD Real Estate Conference.

Seminars: CityUHK, HKU, Texas A&M, SMU, Boston College, University of Toronto Rotman & Mississauga, Purdue University, Rice University, Federal Reserve Board, University of Minnesota, University of Colorado Boulder.

#### 2021

Conferences: Stanford Institute for Theoretical Economics (SITE), NBER Summer Institute Household Finance Conference, NBER Summer Institute Monetary Policy Conference\*, SFS Cavalcade North America, Financial Intermediation Research Society (FIRS), CFPB Research Conference, Society of Labor Economists (SOLE), Southwest Finance Association (SWFA), Royal Economics Society (RES), Swiss Society for Financial Market Research (SGF), Asian Meeting of the Econometric Society, International Association for Applied Econometrics (IAEE), OSU PhD Real Estate Conference, Week-After Conference on Financial Markets and Institution\*.

#### 2020

Conferences: Winter Meeting of the Econometric Society, Atlanta Fed/Princeton Bendheim Conference on Racial Justice and Finance\*, System Applied Microeconomics Conference, System Econometrics Conference\*.

#### 2018

Conferences: International Industrial Organization Conference.

\* Indicates presentation by co-author, <sup>s</sup> indicates scheduled presentation

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|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Conference organization: | 2026 Holden Conference in Finance and Real Estate Program Committee<br>2026 Texas Monetary Conference Co-organizer<br>2026 AREUEA National Conference Program Committee<br>2026 EFA Cavalcade Program Committee<br>2026 SFS Cavalcade Program Committee<br>2026 Georgia Tech-Atlanta Fed Household Finance Conference Program Committee<br>2026 MFA Conference Program Committee<br>2025 EFA Conference Program Committee<br>2025 SFS Cavalcade Program Committee<br>2025 Georgia Tech-Atlanta Fed Household Finance Conference Program Committee<br>2024 EFA Conference Program Committee<br>2024 MFA Conference Program Committee<br>2024 AREUEA/ASSA Program Committee<br>2023 Lone Star Finance Conference Co-organizer<br>2022 SFA Program Committee |
| Referee Service:         | <i>Quarterly Journal of Economics, Journal of Finance, Journal of Financial Economics, Review of Financial Studies, Review of Economic Studies, Journal of Financial and Quantitative Analysis, Management Science, Journal of Monetary Economics, Journal of Urban Economics, Journal of Political Economy: Microeconomics, Real Estate Economics, Journal of Real Estate Finance and Economics, Journal of Housing Economics, Journal of Corporate Finance, Journal of Banking and Finance, Review of Corporate Finance Studies, Journal of Empirical Finance, Journal of Financial Research, Managerial and Decision Economics, Managerial Finance, International Review of Economics &amp; Finance</i>                                                |

### **Honors, Scholarships, and Fellowships:**

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|-----------|----------------------------------------------------------------------------------|
| 2024      | Jensen Prize (First Place), Journal of Financial Economics                       |
| 2024      | UD/Philly Fed Fintech Financial Institutions Conference Best Paper Prize         |
| 2021      | Best Paper Award, OSU PhD Real Estate Conference                                 |
| 2019      | Lab for Economic Applications and Policy (LEAP) grant, Harvard University        |
| 2018      | Certificate of Teaching Excellence, Harvard Bok Center for Teaching and Learning |
| 2018      | Wayfair Datathon, 1 <sup>st</sup> place team, \$20000 prize                      |
| 2016-2021 | Harvard Business School Doctoral Fellowship                                      |
| 2013      | Bernstein Prize in Economics                                                     |

### **Publications:**

Gao, Janet, Livia Yi, and David Zhang. Forthcoming. “Algorithmic Underwriting in High Risk Mortgage Markets”. *Journal of Finance*.

Gerardi, Kristopher, Paul S. Willen, and David Hao Zhang. 2023. “Mortgage Prepayment, Race, and Monetary Policy”. *Journal of Financial Economics* 147 (3):498–524. Editor’s Choice, March 2023. Jensen Prize (First Place), July 2024.

Ishii, Jun, and David Hao Zhang. 2017. “Options Compensation as a Commitment Mechanism in Oligopoly Competition”. *Managerial and Decision Economics* 38 (4):513–525.

Shy, Oz, Rune Stenbacka, and David Hao Zhang. 2016. “History-based versus uniform pricing in growing and declining markets”. *International Journal of Industrial Organization* 48:88–117.

### **Working Papers:**

“*Measuring the Impact of the NAR Settlement*”, with Jefferson Duarte

We empirically examine the effects of the 2024 National Association of Realtors (NAR) settlement on the market for real estate intermediation and document four main findings. First, following the settlement, buyers became two percentage points more likely to forgo formal representation, an effect concentrated in states without prior Buyer's Representation Agreement (BRA) requirements. Second, commission rates show no significant change. Third, the market share of FinTech and discount brokerages remains low. Fourth, transactions increasingly involve seller and buyer agents with higher past volumes and greater experience. Overall, the NAR settlement had only a modest impact on the \$180 billion annual volume of real estate commissions.

“*The Elasticity of Home Purchases to Financing Costs: Evidence from Recent GSE Pricing Changes*”, with You Suk Kim and Feng Liu

Whether home purchases respond to financing costs beyond the mechanical effect of debt-to-income constraints is a key question for macroeconomic models. We study this question using a natural experiment where the government-sponsored enterprises (GSEs) guarantee fee (g-fee) pricing changed differentially across credit score buckets. We find large elasticities of home purchases of 21–25% per 50 bps relative decrease in the fee-adjusted GSE interest rate, driven by tight housing markets. A structural model calibrated to our estimates fits several aggregate elasticities in the literature, and we use it to evaluate recent g-fee adjustments and proposals to improve GSE capitalization.

“*Is There a Puzzle in Underwater Mortgage Default?*”, with Lara Loewenstein, Paul Willen, and Yuxi Yao (**R&R at Journal of Financial Economics**)

A recurring question in the mortgage default literature is why underwater default is rare relative to model predictions. In sharp contrast to the literature, we find that the financial incentives of negative-equity strategic default are significantly lower than previously understood once we incorporate realistic rent dynamics and detailed choice sets of borrowers in a novel quantitative model of mortgage default. We present further empirical results consistent with the unique implications of our model. Our model offers a novel benchmark for interpreting the empirical evidence on mortgage default and reinforces the effectiveness of cash-flow-based interventions.

“*Mortgage Lock-in, Lifecycle Migration, and the Welfare Effects of Housing Market Liquidity*”, with Kristopher Gerardi and Franklin Qian

We develop a novel search and matching model to study the effects of mortgage lock-in across geographic areas and over household lifecycles. We find that younger home buyers are more affected by mortgage lock-in because of a higher baseline level of mobility, disrupting their pattern of moving to higher-quality neighborhoods. We estimate a model of seller-buyers bargaining with endogenous migration across markets. We find that without mortgage lock-in, time on market on average increases by 9% and house prices decline by 10%, with larger effects in high-income areas and superstar cities and larger welfare effects among younger borrowers.

“*Testing for Discrimination in Menus*”, with Paul Willen (**R&R at American Economic Review**)

How should researchers test for differences in the menus of options that people face when given data on choices? In mortgage and labor contexts, we show how intuitively appealing regression-based approaches for assessing differences in menus can lead to misleading and contradictory results. To address this issue, we use pairwise dominance relationships in choices that can be supplemented by restrictions on the range

of plausible menus to define (1) a test statistic for equality in menus and (2) a difference in menus (DIM) metric. We also derive a new procedure for inference on our class of problems. Finally, we apply our methodology to a novel data set linking 2018--2019 Home Mortgage Disclosure Act (HMDA) data to Optimal Blue rate locks. We find evidence for mortgage menu differences by race, particularly among Conforming mortgage borrowers who are relatively creditworthy.

*“Non-Bank Competition and Regressive Mortgage Credit Redistribution”*, with Yongqiang Chu and Tim Zhang

What explains banks' shift towards larger loans following the financial crisis? Using a novel shift-share instrument interacting pre-Basel III market shares with lender-level growth, we show that small-dollar mortgages in areas more exposed to non-bank growth experience lower delinquency rates, higher underwriting quality, and higher denial rates by banks. A 10 percentage point increase in non-bank market share raises bank denial rates by 3.8 percentage points. Mortgage interest rates and loan costs also increase significantly. Regression discontinuity results are consistent with banks' strategic refocusing toward larger (jumbo) mortgages, which results in a reduction in total small-dollar home-purchase mortgage originations for owner-occupied properties.

*“Closing Costs, Refinancing, and Inefficiencies in the Mortgage Market”*

I use a structural model to quantify the cross-subsidization in the US mortgage market due to heterogeneous borrower refinancing tendencies. Actively refinancing borrowers gain up to 4.4% of their loan amount relative to non-refinancing borrowers in expectation. In equilibrium, the presence of borrowers with high refinancing inertia reduces mortgage interest rates particularly on lower upfront closing cost mortgages which have more valuable refinancing options. As a result, actively refinancing borrowers refinance excessively relative to a perfect information, no cross-subsidization benchmark, an effect that accounts for 36% of the rate and term refinances for 2013--2019 new purchase originations and generates deadweight losses due to administrative resource costs. Alternative contract designs can simultaneously reduce transfers and increase total welfare.

*“The Cost of Being Underbanked: Racial Disparities in Access to PPP Loans and its Equilibrium Implications”*, with Jeffrey Wang

Many government support programs for small businesses are designed to pass through banks and credit unions. However, this poses barriers for minority communities that are less connected to financial institutions for obtaining this support. Using the latest program for supporting small businesses, the Paycheck Protection Program (PPP), as an example, we show that there was a large disparity in the density of PPP enrolled lenders by racial composition of the neighborhood. This difference is both due to a lower density of lenders in those neighborhoods in general, and by the fact that the banks and credit unions that do operate there are smaller, are less likely to have previous relationships with the Small Business Administration, and are less likely to enroll in the program. More heavily Black neighborhoods have significantly lower take-up of PPP loans particularly in lower population (more rural) areas where this disparity is most salient. Through an instrumental variables analysis, we show that the intensive margin of access to enrolled lenders can explain about 35% of the racial disparity in take up within the relevant areas. Our results suggest that government programs that provide "support through banks" can have undesirable distributional implications.

*“Do Judge-Lawyer Relationships Influence Case Outcomes?”*, with Tianwang Liu (**R&R at Economic Inquiry**)

We examine whether law school alumni relationships between the lawyers and judges affect case outcomes. We show that in the context of medical malpractice lawsuits in Florida, the plaintiff lawyer sharing the same law school as the judge increases the chances of recovery by 2%. Furthermore, the effect is confined to younger lawyers who see a 4% increase in the likelihood of recovery from having been to the same law school as the judge, and is absent in older lawyers. We interpret our results as evidence that

lawyers gain school-specific human capital from their law schools which helps in their interactions with judges that graduated from the same school, and that this school-specific human capital become less important further on in the lawyers' careers.

*“Semi-Parametric Estimation of Counterfactuals in Dynamic Discrete Choice Models”*

I develop a new method for estimating counterfactuals in dynamic discrete choice models, a widely used set of models in economics, without requiring a distributional assumption on utility shocks. Applying my method to the canonical Rust (1987) setting, I find that the typical logit assumption on utility shocks can lead the researcher to conclude that the agent's counterfactual choice probabilities are much more sensitive to policy changes than what a semi-parametric model would suggest. Therefore, my method may be useful to applied researchers in generating policy counterfactuals that are robust to such distributional assumptions.

**Citizenship Information:**

Canadian citizen, US Permanent Resident